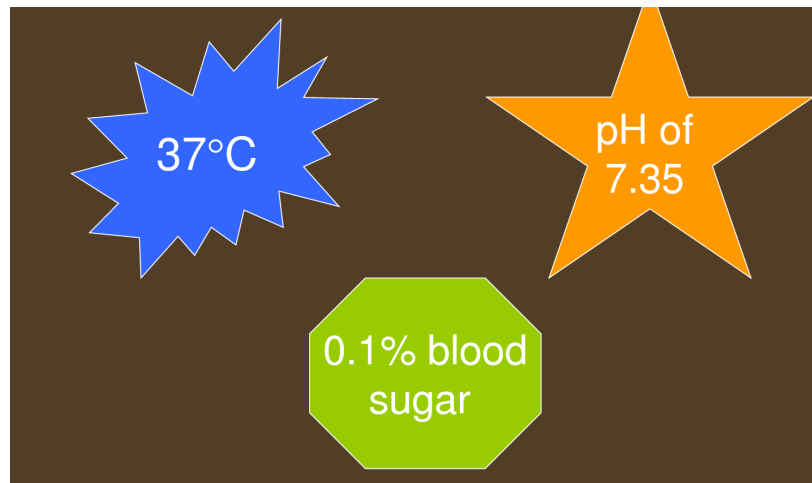


Homeostasis

Lecture 17



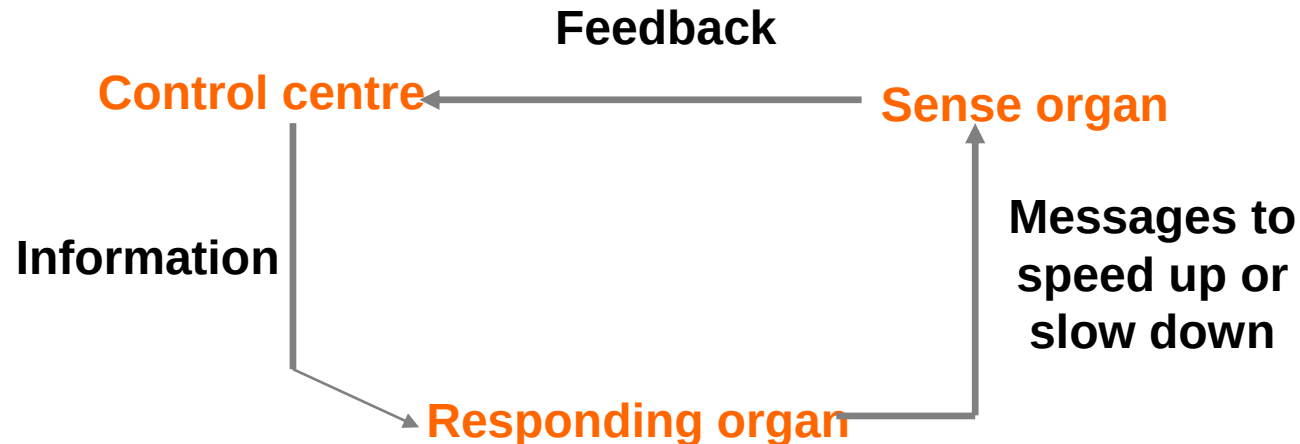
Homeostasis Control of Homeostasis

The maintenance of a **constant environment** in the body is called **Homeostasis**

Homeostasis is controlled by a feedback mechanism.

The feedback mechanism consists of **three parts**:

1. A sense organ
2. A control center (usually the brain)
3. A responding organ

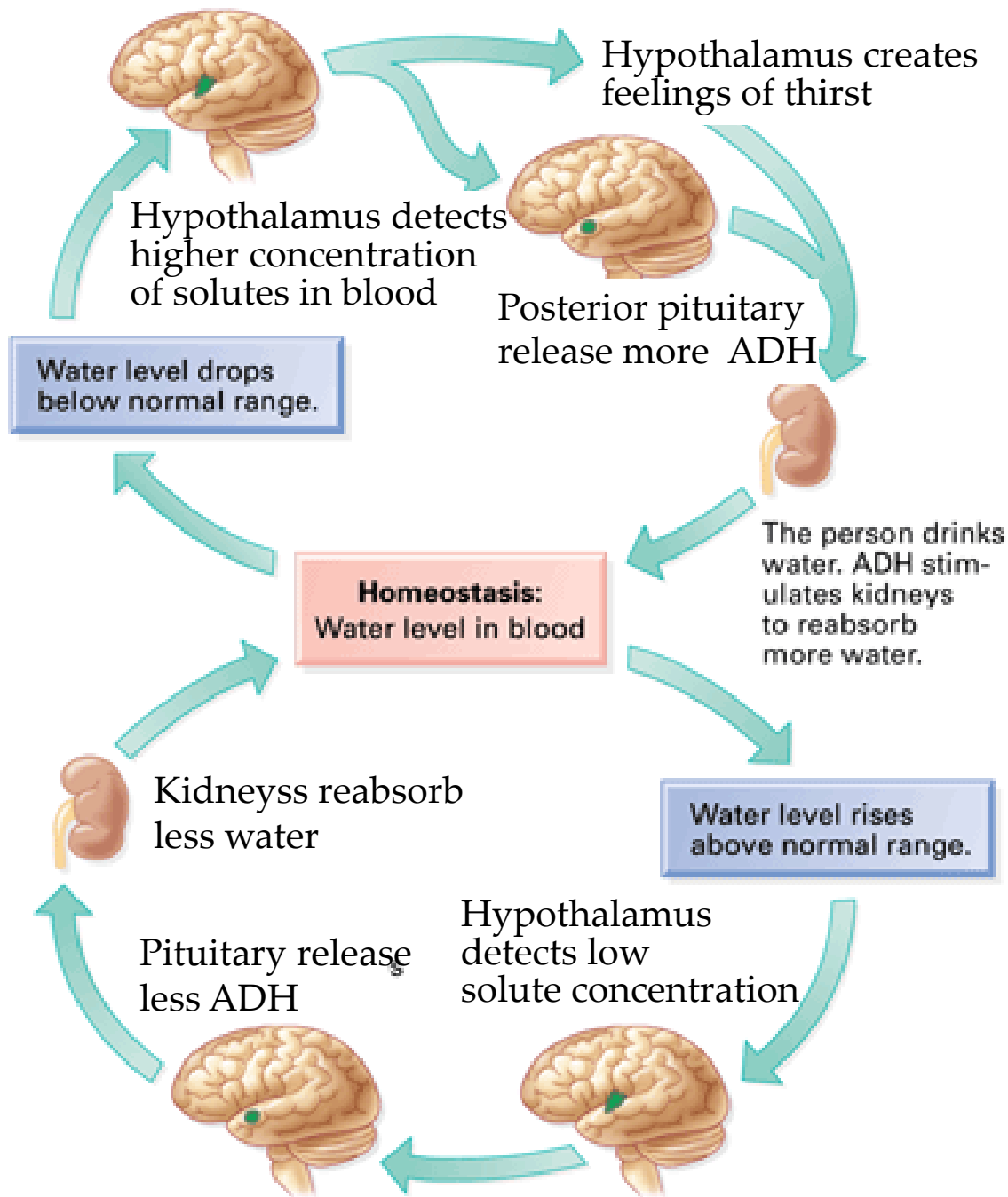


The 3 parts of a typical feedback mechanism

Organs of Homeostasis

- Lungs and respiratory tract
 - Heart and circulatory system
 - Liver, pancreas and gastrointestinal tract
 - Kidneys
 - Musculoskeletal system
 - Skin
 - Nervous system
 - Endocrine glands
- **Body Temperature**
 - **Blood pressure**
 - **Blood pH**
 - **O₂ and CO₂ concentration**
 - **Osmoregulation-Water balance**
 - **Blood glucose**

Water Homeostasis (osmoregulation)



When H_2O level drops →
Hypothalamus creates thirst &
posterior pituitary releases **more**
Anti Diuretic Hormone (ADH) →
More H_2O intake & more H_2O are
reabsorbed in the kidney.

When H_2O levels are high →
Hypothalamus detects low solutes
& pituitary releases **less ADH** →
Less H_2O are reabsorbed in the
kidney.

Insulin secretion mechanism:

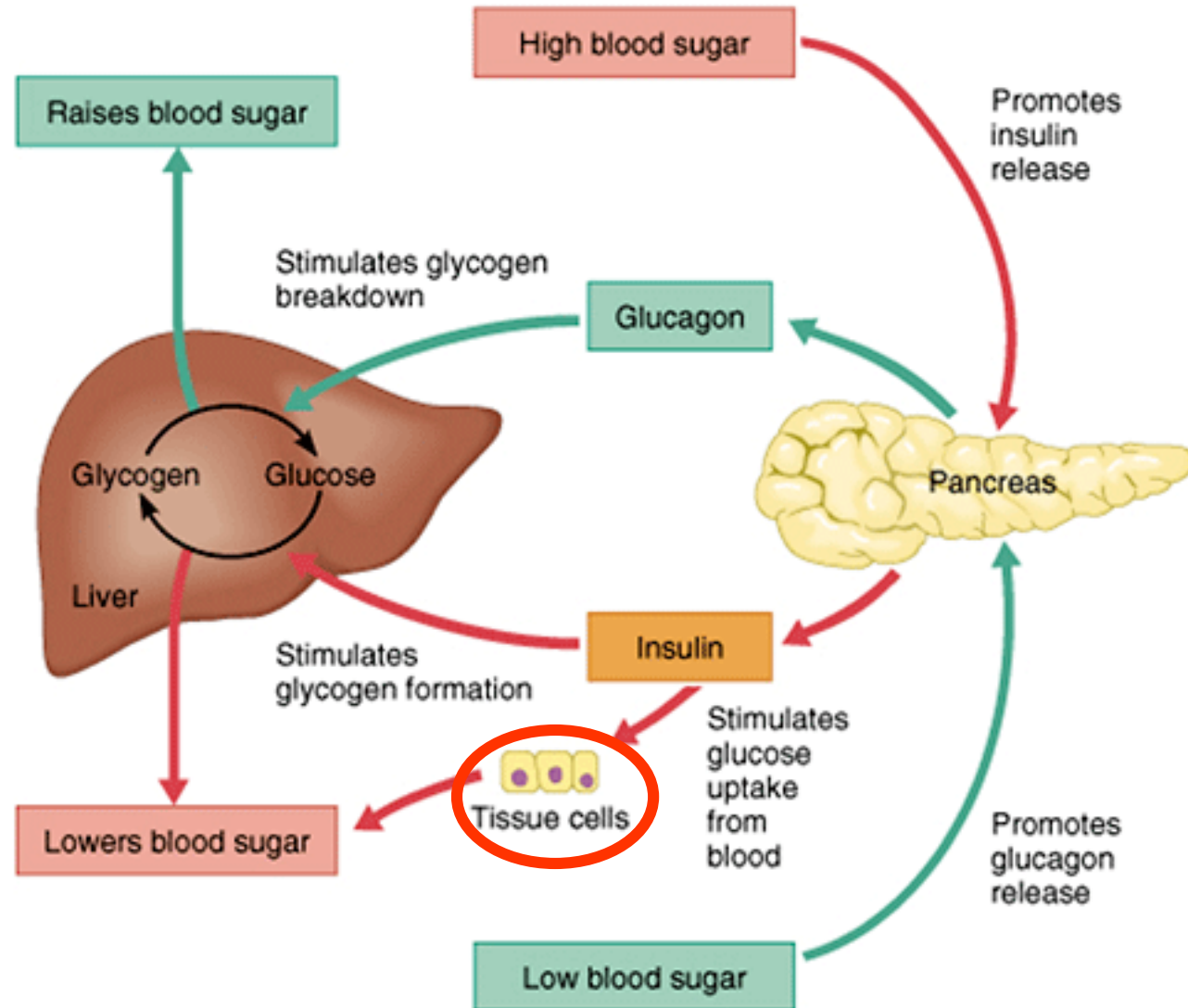
- **High blood glucose** after meals (also amino acids, fatty acids)
- Glucose enters **β -cells of the pancreas** (in the islets of Langerhans) via **GLUT2** (Glucose transporter 2).
- Metabolism $\rightarrow$ $\uparrow$ ATP/ADP ratio $\rightarrow$ closure of ATP-sensitive K^+ channels.
- Depolarization $\rightarrow$ opening of voltage-gated Ca^{2+} channels.
- **Ca^{2+} influx (enter into β -cells) triggers insulin granule exocytosis** into blood $\rightarrow$ Liver and tissues uptake glucose.

Role of Glucagon:

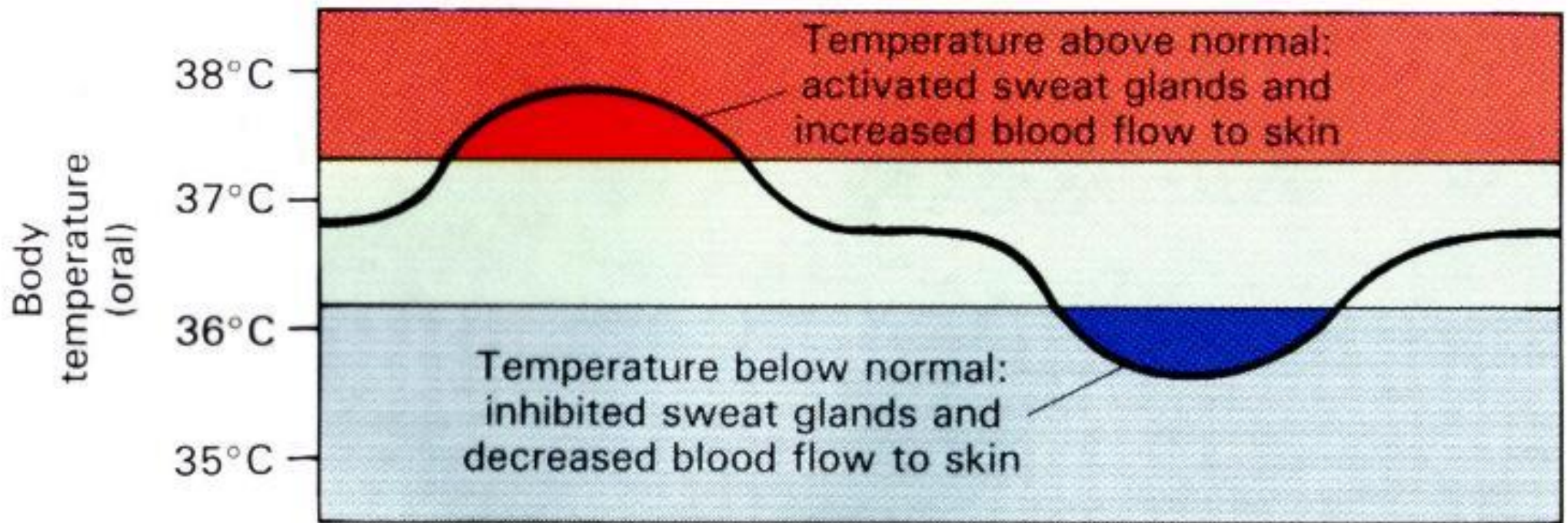
- Low glucose $\rightarrow$ α -cell detects $\rightarrow$ Ca^{2+} influx $\rightarrow$ Glucagon release $\rightarrow$ Liver makes/releases glucose $\rightarrow$ Blood glucose restored

Glucose Homeostasis

[The body's ability to keep blood glucose levels within a narrow range (about **70–110 mg/dL fasting**), despite changes from eating, fasting, or exercise]



Temperature Regulation

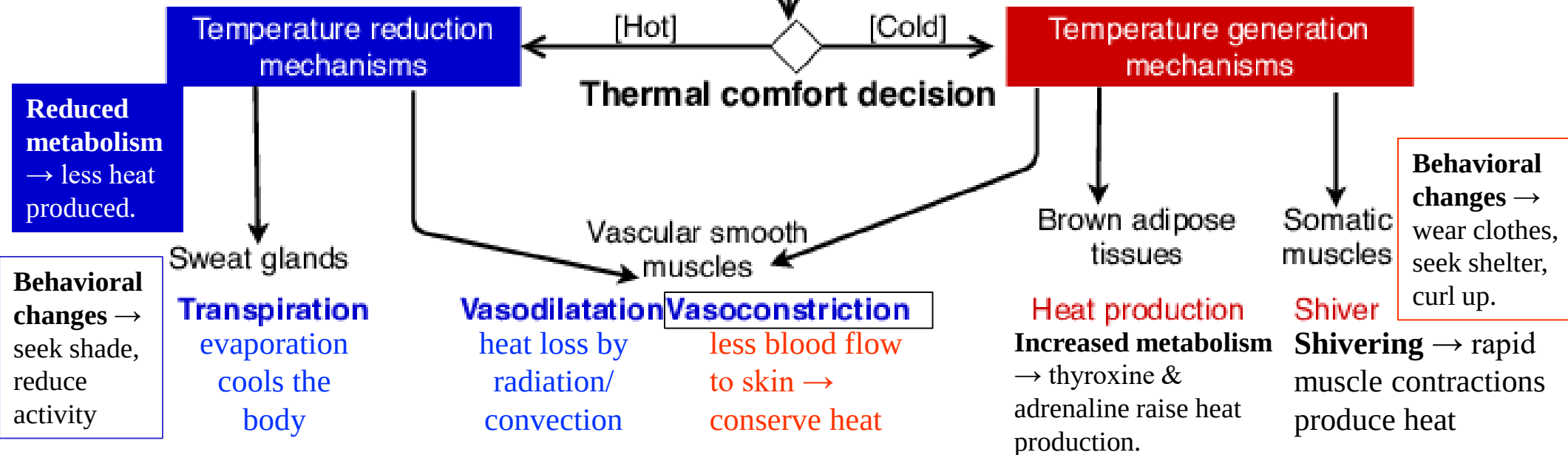
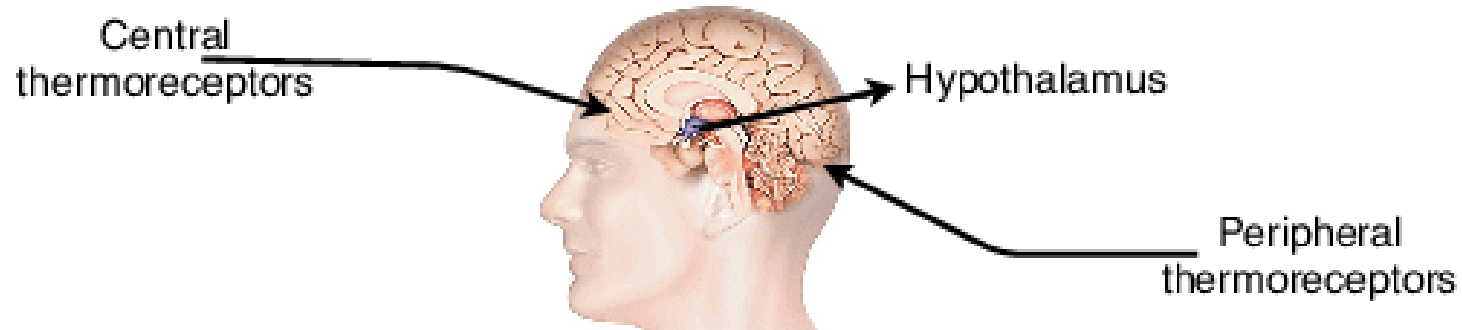


If **too high** → trigger heat-loss mechanisms.
If **too low** → trigger heat-gain mechanisms.

Temperature homeostasis (Thermoregulation)

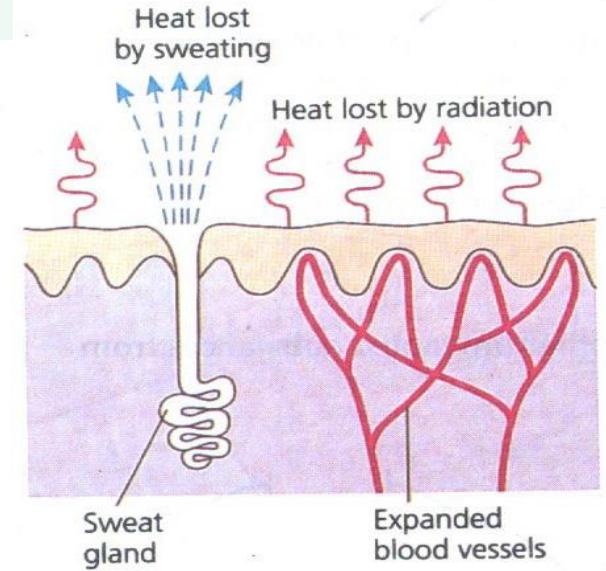
An area in the Hypothalamus called the “**Thermo-Regulatory Center (TRC)**” has the responsibility of **maintaining the body’s core temperature** (it is called body’s thermostat).

It can respond to extremely small changes in blood temperature.

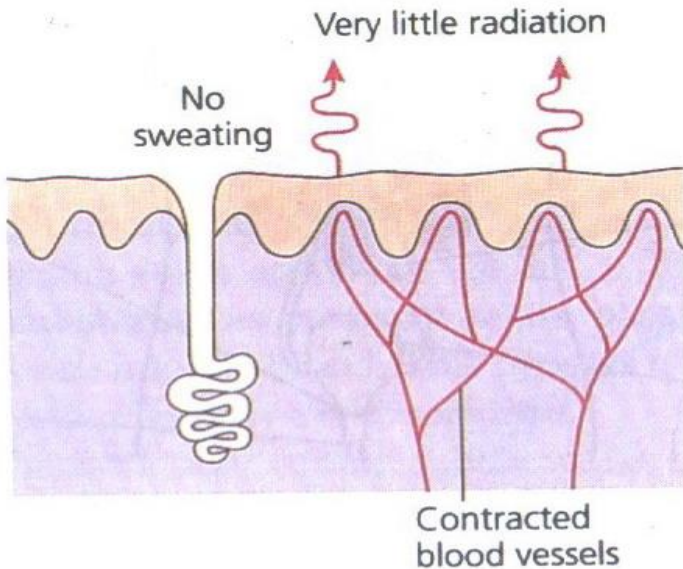


Temperature Regulation

- When we get **too hot**, we **sweat a lot**. The sweat evaporates and cools us down.
- The blood vessels below our skin **expand**. So, a **lot of blood flows** near the body surface and **loses heat**.



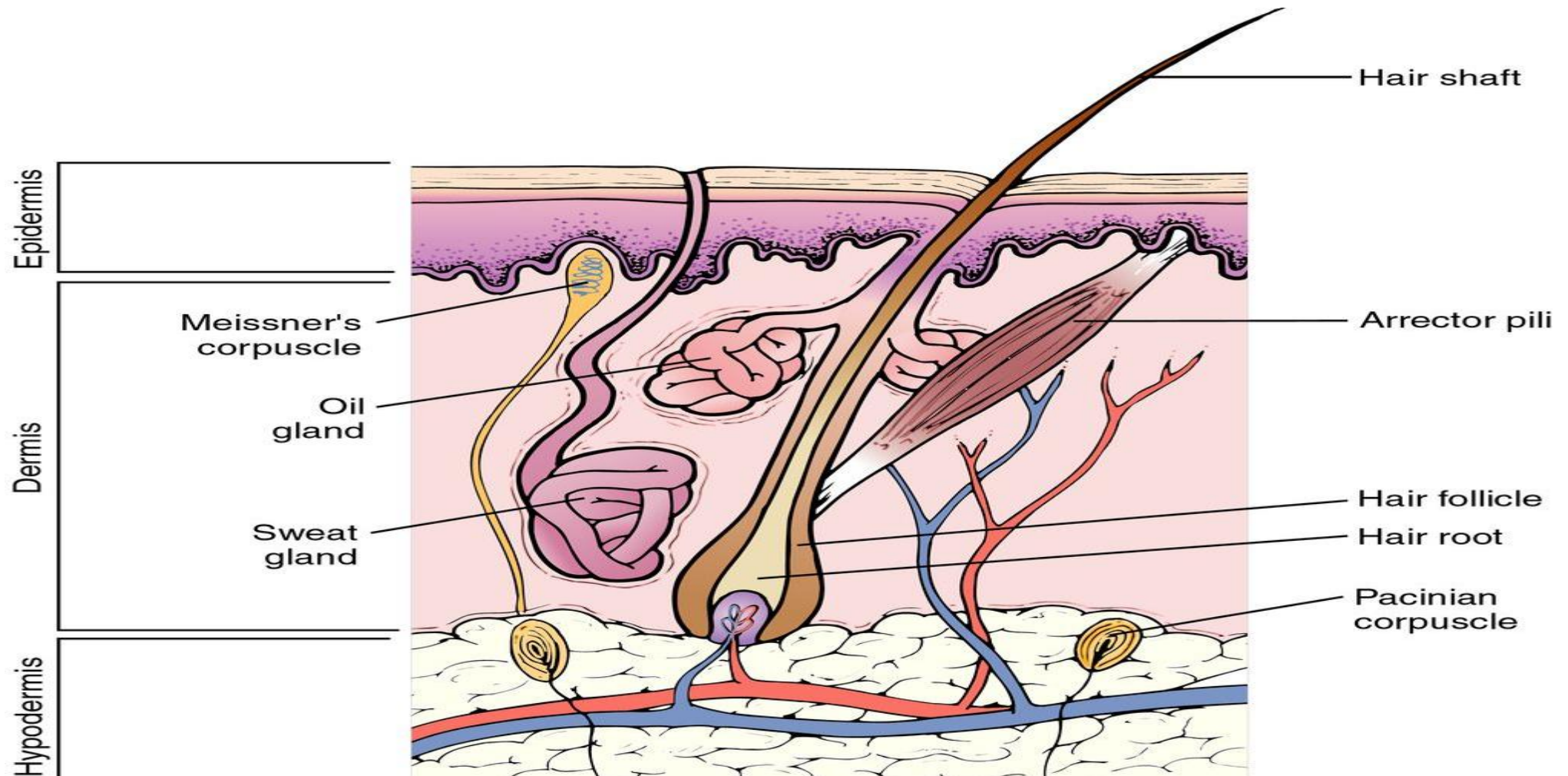
During Hot Weather



During Cold Weather

- When we get **too cold**, we stop sweating, blood vessels below the skin contract.
- So only a **little blood flows near the surface** and loses heat.
- **Muscles start rapid, jerky movements** which we call **shivering**. This produces extra heat, which warms up the body.

Temperature Regulation

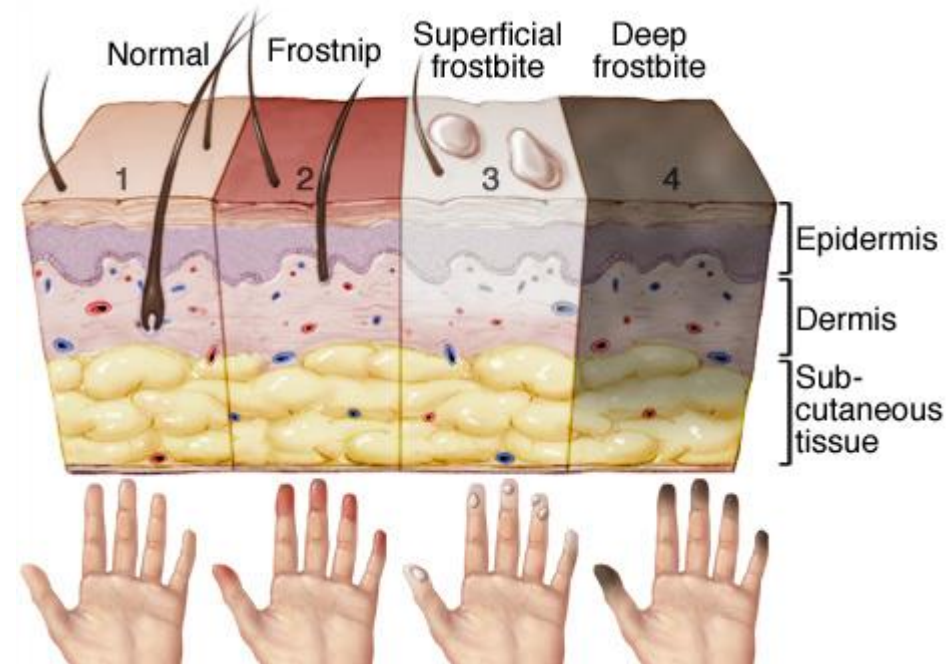


Fever and hyperthermia

1. Fever occurs when the *core temperature of the body* is raised above normal steady state levels. The body reacts as if it is too cold, while the temperature rises up to the **new higher set point**. Fever attacks imply shivering combined with vasoconstriction, headache, body pains, and general discomfort.
 2. Fever results from the action of *endogenous pyrogens* on the *hypothalamic heat control centre* [they **increase the set point for the core temperature via prostaglandins**].
 3. Exogenous pyrogens from microbes cause these *endogenous polypeptides* to be released from the defense cells of the body. [Antipyretic drugs inhibit cyclo-oxygenase activity, hereby interfering with the synthesis of prostaglandins and thromboxanes].
 4. Following an attack of fever, vasodilatation and sweat evaporation reduces the core temperature.
- *Pyrogen is a substance, that causes a rise in temperature in a human or an animal*
 - *Prostaglandin (PG) is a potent hyperthermic agent*

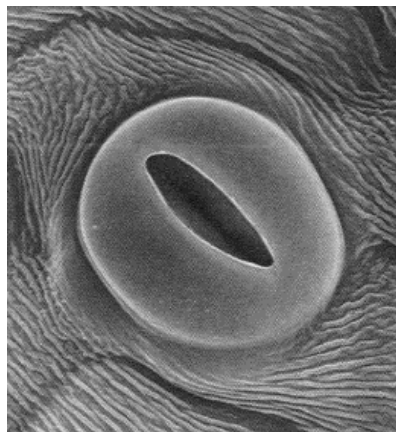
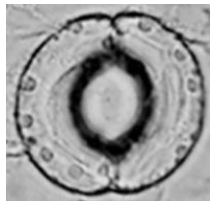
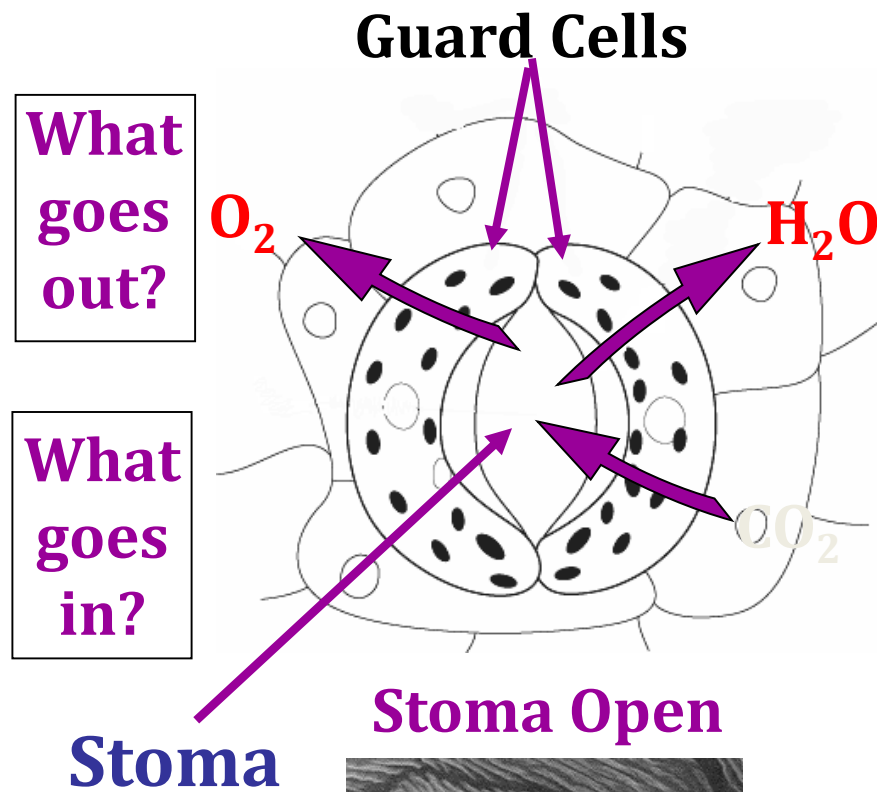
Excessive Heat Loss Can cause Hypothermia and Frostbite

Hypothermia (colder than normal body temperature) and frostbite can happen to anyone who is not properly protected from the cold.



- Hypothermia occurs when the body **temperature falls below 95 degrees F (35 degrees C)**. The normal body temperature is 98.6 degrees F. In the early stages, this may cause memory loss, confusion and shivering. Eventually, a low body temperature may cause cardiac arrest and death.
- The best way to prevent hypothermia is to protect the body from the cold.

Homeostasis in Plants: Function of Stomata

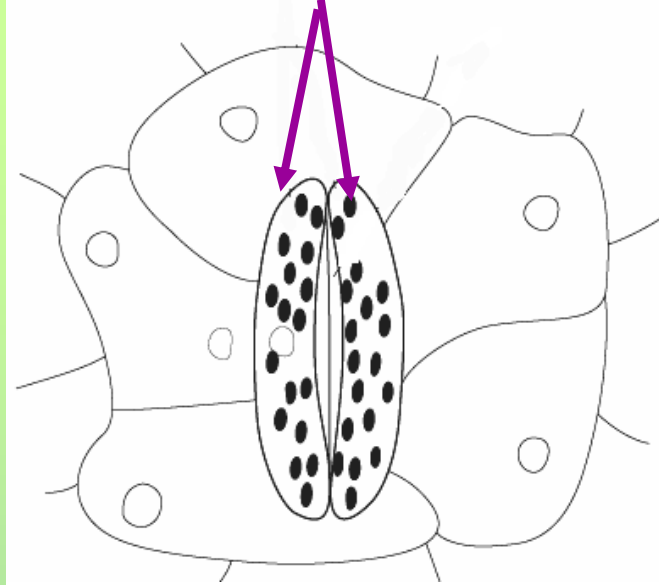


- What process involves using CO_2 and H_2O releasing O_2 as a waste product?
- Photosynthesis
- What is the plant using this process to make?
- Carbohydrates-glucose
- If the plant needs *water* for photosynthesis, why is *water* coming out of the stoma?

Function of Guard Cells

- These stomata (leaf openings) naturally allow water to evaporate out.
- Why would the plant close stomata with guard cells?
- Prevent **excess** water loss through transpiration. (conserve water)
- So what is the point of having stomata?
- Allow gas exchange for photosynthesis

Guard Cells



Stoma Closed

